

Indirect Attack

—Side Channel, Power, Timing

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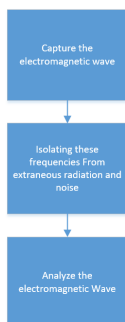


Introduction

Side-channel attack is any attack based on information gained from the physical implementation of a cryptosystem. For example, timing information, power consumption, electromagnetic leaks or even sound can provide an extra source of information, which can be exploited to break the system. Some side-channel attacks require technical knowledge of the internal operation of the system on which the cryptography is implemented, although others such as differential power analysis are effective as black-box attacks.

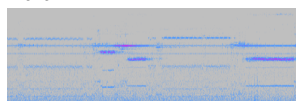
Side Channel

Process

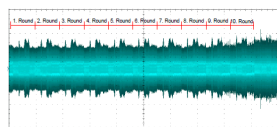


Electromagnetic attacks are side-channel attacks performed by measuring the electromagnetic radiation emitted from a device and performing signal analysis on it. Electronic devices create some small magnetic fields when in use. These magnetic fields can unintentionally reveal information about the operation of a device if not properly designed.

Data



A spectrogram showing RSA encryption and decryption.

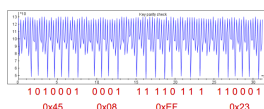


A spectrogram showing AES encryption.

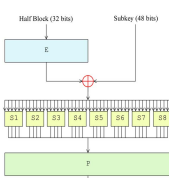
Power

Power analysis is a form of side channel attack in which the attacker studies the power consumption of a cryptographic hardware device (such as a smart card, tamper-resistant "black box", or integrated circuit).

Simple power analysis (SPA): involves visual examination of graphs of the current used by a device over time.



Differential power analysis (DPA): involves statistically analyzing power consumption measurements from a cryptosystem.



The attack exploits biases varying power consumption of microprocessors or other hardware while performing operations using secret keys.

How to Prevent

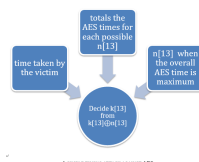
SPA:
Avoid secret intermediates or keys
Avoid branching operations
DPA:
Reduce signal sizes
Introducing noise into power consumption measurements
Nonlinear key update procedures

Other Attack Mode

- Acoustic Cryptanalysis — attacks that exploit sound produced during a computation (rather like power analysis).
- Differential Fault Analysis — in which secrets are discovered by introducing faults in a computation.
- Data Remanence — in which sensitive data are read after supposedly having been deleted.
- Row Hammer — in which off-limits memory can be changed by accessing adjacent memory.

Timing

A timing attack is an example of an attack that exploits the **data-dependent** behavioral characteristics of the implementation of an algorithm rather than the mathematical properties of the algorithm itself.



Timing attacks are easier to mount if the adversary knows the internals of the hardware implementation, and even more so, the crypto system in use.

- As the amount of noise in the side channel increases, the adversary needs to collect more measurements
- Run in an exactly constant amount of time, independently of secret values
- CPU manufacturers should thoroughly document the performance of their chips



SOME METHODS TO PREVENT TIMING ATTACKS

Conclusion

- Side-channel attack is a new field now and is mainly used in algorithm design and circuit design. We need to develop the effective simulation tools of side-channel attack.
- The attacks are easy to implement, have a very low cost per device, and are non-invasive, making them difficult to detect.
- A correct implementation of a strong protocol is not necessarily secure. The protection need to be well improved in the future.

Reference

- Bernstein, Daniel J., Cache-timing attacks on AES, 2005.
- GB/T 7714 Martinasek Z, Zeman V, Trasy K. Simple electromagnetic analysis in cryptography[J]. International Journal of Advances in Telecommunications, Electrotechnics, Signals and Systems, 2012, 1(1): 13-19.
- P. Kocher, J. Jaffe, B. Jun, "Differential Power Analysis," technical report, 1998; later published in Advances in Cryptology - Crypto 99 Proceedings, Lecture Notes In Computer Science Vol. 1666, M. Wiener, ed., Springer-Verlag, 1999.

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Security